GlycTwin-AI Presentation Deck

Happiest Health Reimagining & Reforming Healthcare in India Summit - 2026

15-Slide Presentation Deck Summary (Complete PPTX available at docs/presentation.pptx)

1. Executive Summary: Proactive Precision Digital Twin for T2D in India
2. Problem Statement: South Asian Metabolic Phenotype & Postprandial Spikes
3. Dual Data Fusion: Merging Static EHR + Dynamic 5-min IoT Sensor Stream
4. Feature Engineering: Glucose Velocity, Active COB/IOB Decay, Circadian Encodings
5. AI Architecture: Physics-Informed Bergman Minimal Model + Fused LightGBM Core
6. Validation Metrics: RMSE 6.68 mg/dL, R^2 0.9503, Clarke Grid Zone A 99.7%
7. Early Warning Engine: 2-Hour Advance Hyperglycemia Spike ROC-AUC 0.9879
8. Clinical UI Dashboard: Real-time CGM, Radar Twin State, SHAP Feature Attribution
9. 'What-If' Counterfactual Sandbox: Interactive Meal, Walk & Insulin Titration Sim
10. CDSS Engine: Actionable Evidence-Based Clinical Recommendations
11. Privacy & Governance: Complete DPDP Act (India) & HIPAA Sandbox Compliance
12. Technical Stack: Python, LightGBM, FastAPI, Chart.js, Tailwind CSS
13. Scalability Roadmap: ABDM / ABHA Integration & Cardio-Renal Digital Twins
14. Team & Incubator Details: IISc / Top University Incubator Track
15. Open Source MIT License & Public GitHub Repository